

LUU DINH MAC

DevOps / Infrastructure Engineer

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PROFESSIONAL SUMMARY

Experienced Infrastructure & DevOps Engineer specializing in Linux, virtualization, storage, and cloud infrastructure. Proven ability to troubleshoot critical production systems, automate operations, and maintain high-availability environments. Hands-on experience building and operating self-hosted Kubernetes environments with GitOps, CI/CD automation, observability platforms, and cloud migration to CMC Cloud.

WORK EXPERIENCE

Infrastructure Engineer | Saigon Precision — Misumi Group

Mar 2026 – Present • **CURRENT**

- Migrated production workloads from on-premise VMware to CMC Cloud, ensuring service continuity and minimal disruption.
- Designed and automated deployment of enterprise monitoring platforms using Ansible, Docker Compose, Zabbix, and Grafana for consistent, repeatable provisioning.
- Built custom dashboards, metrics, triggers, and alerting workflows — integrated with Microsoft Teams via webhook pipelines for real-time incident management.
- Implemented real-time network topology and bandwidth monitoring using Grafana Network Weathermap.
- Administered VMware virtualization environments supporting Oracle Database and Microsoft SQL Server production workloads.
- Managed enterprise services: firewall, VPN, Active Directory, virtualization platforms, and internal network services.
- Operated Veeam Backup & Replication with offsite S3-compatible object storage; performed recovery testing and retention policy management.
- Collaborated with dev teams to provision infrastructure, support deployments, and resolve environment issues across dev/test/prod.

Systems Deployment Engineer | Thi Thien Solutions Technology Corporation

Nov 2023 – Mar 2026

- Configured SQL Server 2022 Failover Cluster Instance (FCI) on Windows Server 2022 across 2-node clusters for enterprise HA.
- Implemented iSCSI Direct Attach, MPIO, and Storage Tiering (SSD/HDD) on HPE MSA 2060 for optimized performance and redundancy.
- Recovered ESXi 8 root access for Post Office General Hospital — restored critical hospital systems after 10-day outage within 2 hours.
- Recovered ESXi 7 credentials for Vietsovetro Resort Ho Tram with downtime under 1 hour, restoring full operations.
- Resolved HA synchronization issues on HPE SimpliVity production cluster for Saigon Giai Phong Newspaper.
- Diagnosed complex hardware/software issues for SAN storage systems (Dell, IBM, 3PAR, MSA), maintaining 99.9% uptime.
- Supported hospital (Hung Vuong) with VMware ESXi 8 setup, Oracle Linux installation, and Oracle Database configuration.
- Installed CUDA drivers and configured TensorFlow, PyTorch, and Jupyter Notebook with GPU support for AI/ML workloads.

Freelance System Engineer | Kubernetes & OpenStack Deployment

Apr 2025 & Oct 2025

- Troubleshooted Kubernetes cluster issues preventing worker nodes from joining; restored production workloads by redeploying existing manifests (Apr 2025).
- Investigated OpenStack deployment failures (Kolla-Ansible) — resolved service startup errors and dependency issues, bringing all services online (Oct 2025).

IT Systems Specialist | Delta Engineering Corporation (now DEC Engineering JSC)

Sep 2020 – Apr 2022

- ▶ Deployed and operated OpenStack, Proxmox cluster, and CEPH distributed storage for private cloud infrastructure.
- ▶ Implemented HA architectures for MariaDB and PostgreSQL using HAProxy, Galera Cluster, and Patroni for automatic failover.
- ▶ Built and operated Private Cloud with Nextcloud integrated with ADFS SSO for Microsoft Office 365.
- ▶ Deployed centralized monitoring using Zabbix and Grafana; configured firewalls (Sophos, Fortinet, pfSense).
- ▶ Automated routine sysadmin tasks with Bash and Python, reducing manual effort by 80%.
- ▶ Managed internal IT: user account provisioning, Active Directory, Microsoft 365, and domain controller administration.

System Operation | Berjaya Gia Thinh Investment Technology JSC (Vietlott)

Jan 2019 – Dec 2019

- ▶ Operated and monitored server infrastructure for Vietlott lottery systems; managed daily/weekly tape backups.
- ▶ Operated OSS (Online Sell Server) on OpenVMS; managed system start/close cycles during lottery draws.

NOC Operator Trainee | Online Mobile Services JSC (MoMo)

Jan 2018 – Aug 2018

- ▶ Monitored server infrastructure and services using Nagios and Grafana; logged and resolved anomalies to ensure platform availability.

PERSONAL PROJECT

Personal Blog Platform & Self-Hosted Cloud-Native Infrastructure

2024 – Present | blog.luumac.io.vn | github.com/luudinhmac/myweb-profile-blog

Designed, built, and operated a production-grade self-hosted platform following Smart Server / Lean Client architecture and Documentation-as-Code (DaC) philosophy — treating all infrastructure decisions, runbooks, and incident reports as version-controlled artifacts alongside application code.

Infrastructure & Kubernetes

- ▶ Provisioned and hardened bare-metal Linux servers using Ansible: SSL/TLS management, DNS, Traefik Ingress with HostNetwork mode, server hardening.
- ▶ Deployed and managed Kubernetes workloads via Helm and ArgoCD GitOps — full declarative cluster state through a dedicated infrastructure repository.
- ▶ Implemented Sealed Secrets for secure secret management in K8s, preventing sensitive data exposure in Git.
- ▶ Configured HPA and resource policies across namespaces for stable multi-workload operation; used Cloudflare R2 as cost-efficient object storage backend.

CI/CD & DevSecOps

- ▶ Built GitLab CI/CD pipelines implementing Build Once, Deploy Many: automated build → Trivy vulnerability scan → Docker Hub push → yq image tag mutation → ArgoCD auto-sync.
- ▶ Integrated deployment gates blocking promotion of container images with critical CVEs, enforcing DevSecOps at the pipeline level.

Backup & Disaster Recovery

- ▶ Automated Kubernetes workload backup with Velero and daily etcd snapshots via CronJobs — all stored in Cloudflare R2.
- ▶ Authored Disaster Recovery runbooks: Prisma migration lock recovery, Traefik 403 misconfigurations, physical VPS rebuild, and deep etcd restoration.

Observability & Documentation

- ▶ Full observability stack with Prometheus + Grafana; real-time alerting to Telegram and Microsoft Teams.
- ▶ Maintained structured K8s Incident Log with RCA for real production issues (Velero/Kopia CPU spikes, Next.js hydration failures, standalone proxy freezes).
- ▶ Applied Documentation-as-Code: all ADRs, deployment standards, CI/CD workflows, troubleshooting journals, and operational runbooks are version-controlled in the repository.

Stack: Kubernetes · Docker · Helm · ArgoCD · GitLab CI/CD · Traefik · Ansible · Sealed Secrets · Velero · Prometheus · Grafana · Next.js · NestJS · TypeScript · PostgreSQL · Redis · Cloudflare R2

TECHNICAL SKILLS

Operating System	Linux (Ubuntu, CentOS, Debian), Windows Server, Oracle Linux
Networking & Security	TCP/IP, DNS, DHCP, SNMP, VPN, HA · Firewall (Sophos, Fortinet, pfSense) · Wazuh, CVE Patching

Virtualization	VMware ESXi/vCenter/vSAN · OpenStack (Kolla-Ansible) · Proxmox · KVM · CEPH
Containerization	Docker · Kubernetes · Helm · ArgoCD · Sealed Secrets · Harbor · Rancher · Velero
CI/CD & Automation	GitLab CI/CD · Jenkins · Ansible · Terraform · Trivy · SonarQube
Scripting	Python · Bash
Monitoring & Logging	Prometheus · Grafana · Zabbix · Nagios · ELK Stack
Cloud & Storage	CMC Cloud · AWS (EC2, S3, IAM, VPC) · Cloudflare R2 · HPE MSA · IBM SAN · Dell SAN
Web & App Server	Nginx · Apache · HAProxy · Traefik
Load Balancing & DB HA	HAProxy · Galera Cluster · Patroni (PostgreSQL) · Keepalived
Version Control	Git · GitLab · GitHub

EDUCATION

Bachelor's Degree in Information Technology | HCMC University of Technology and Education
2014 – 2019

CERTIFICATIONS

- DevOps & DevSecOps — DevOpsEdu.vn (2024)
- Practical Kubernetes — DevOpsEdu.vn (2024)
- DevOps on AWS (EC2, IAM, VPC, S3) — DevOpsEdu.vn (2025)
- DevOps for Business — DevOpsEdu.vn (2025)
- CCNA — VNPRO (2017)

Continuous Learning: Actively pursuing online courses and certifications in DevOps, Kubernetes, AWS, and Terraform.